Ahmed Alabdouli

Contribution to phase 3 of the project

File content

•

• 2.8 AI Resource Matching and Ranking (FR-8)

• 2.9 AI Asset Classification (FR-9)

• 2.10 Sustainable-Action Recommendations (FR-10)

• 2.11 AI Assistant (FR-11)

• 2.14 General AI Requirements

• 3.1 Performance Requirements

• 3.4 Software Quality Attributes

• UC-9: Review and Override AI Matches

• UC-10: Classify a New Asset with AI Suggestions

• UC-11: Ask the AI Assistant

• Appendix E: Stakeholder Interviews

• E.1 Facilities Management and Asset Coordinator

• E.2 Procurement and Finance

• E.3 Maintenance and Sustainability

• E.4 IT and System Administrator

• E.5 Department Representatives

• E.6 University Management

2. System Features

2.8 AI Resource Matching and Ranking (FR-8)

2.8.1 Description and Priority

This feature compares a department's request with available campus assets. It understands the meaning of the request as well as the recorded details. It gives the Procurement Team a ranked list, a compatibility score and reasons for each match. The selected asset still follows the normal release, approval and reservation process (FR-4).

Priority: High. This feature develops Phase 2 requirement FR-8. AI-based matching, ranking, explanation and user review are part of the minimum prototype [1].

2.8.2 Stimulus/Response Sequences

1. A Procurement Team member opens a request they are allowed to review and asks for matches. The system checks the request details and available assets.

2. The system ranks suitable assets and shows their scores, confidence, permitted cost details, repair needs and available sustainability estimates.

3. The Procurement Team member accepts a suggestion or chooses another suitable asset. The system records the choice and sends it through the normal release and approval process (UC-4).

4. If details are invalid, no match is found or AI fails, the system explains the problem and offers correction or standard search.

2.8.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-8.1	The system shall match the meaning of a request to asset descriptions and specifications, even when different words are used.	Essential	UC-9	A test request for a "workstation for office tasks" finds a "desktop computer" that meets the required specifications.
FR-8.2	The system shall check required request details and exclude assets that fail access, availability, safety or essential specification checks, including assets under maintenance or pending end-of-life action (FR-5.9, FR-5.27), unsafe assets (SAF-5) and hazardous assets without a safety review (SAF-1).	Essential	UC-9	Missing required details are identified. Unavailable, restricted, unsafe or incompatible test assets do not appear as suitable matches.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-8.3	The system shall consider category, purpose, specifications, quantity, current and requested condition, location, urgency, required date, transfer cost and repair cost.	Essential	UC-9	Each factor shows its value or unknown. The system does not relax a required condition or delivery date without the proper request change.
FR-8.4	The system shall give each suitable match a compatibility score from 0 to 100 and explain its main strengths and limitations.	Essential	UC-9	Every match has a score within the range, a separate confidence label and reasons based on the records. Missing required details prevent scoring.
FR-8.5	The system shall rank suitable assets using documented rules for suitability, urgency, transfer cost, repair needs, waste reduction and estimated carbon reduction. The ranking shall not use the requesting department's name or size as a factor (LEG-9).	Essential	UC-9	Results identify the rule version and explain the main trade-offs. A high score never overrides safety or essential request requirements. Changing only the requesting department does not change the ranking.
FR-8.6	The system shall clearly show when an asset meets only part of the requested quantity. A revised or split request shall be confirmed by the requesting Department Representative and approved under FR-4 before any quantity is reserved.	Essential	UC-9	If five items are requested and only three are available, the result says partial match. The system cannot reserve five or silently change the request to three.
FR-8.7	The system shall let the Procurement Team member accept, reject or choose another suitable match, and record a reason for rejecting or changing the suggested choice.	Essential	UC-9	The record includes the selected asset, original suggestion, reviewer, time and reason. Choosing another asset still requires the usual approvals.
FR-8.8	The system shall recheck availability, quantity, permissions and request details just before sending the chosen asset to the release and approval process (FR-4).	Essential	UC-9	If the asset is no longer available, confirmation is blocked and updated options are shown. The same available units are never offered to two requests at confirmation.
FR-8.9	The system shall explain when no suitable match is found and offer standard search or a request change.	Essential	UC-9	Tests with no match or an AI failure keep the original request and offer a usable alternative. The system does not recommend an unsuitable asset.

2.9 AI Asset Classification (FR-9)

2.9.1 Description and Priority

This feature helps the Asset Coordinator register or update an asset. It suggests details from the title, description, specifications, photos and documents provided. The Asset Coordinator checks the suggestions before they are saved.

Priority: High. This feature develops Phase 2 requirement FR-9. AI-assisted classification is part of the minimum prototype [1].

2.9.2 Stimulus/Response Sequences

1. The Asset Coordinator enters a title and description, adds available specifications and files, and asks for classification suggestions.

2. The system suggests a category, subcategory, tags, possible material type and description. It shows confidence, reasons and any missing details.

3. The Asset Coordinator accepts or corrects the suggestions. The system checks the record and saves only confirmed values.

4. If information conflicts, a file cannot be read or AI is unavailable, the system explains the issue and lets the Asset Coordinator complete the record manually.

2.9.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-9.1	The system shall use available titles, descriptions, specifications, images and documents to suggest asset details, and identify any source it cannot read.	Essential	UC-10	A test asset with text, an image and a document shows which sources were used. Unreadable or unsupported sources are clearly identified.
FR-9.2	The system shall suggest a category and, where relevant, a subcategory from the approved category list.	Essential	UC-10	Every suggestion uses an approved category. If no suitable category can be identified, the system asks for manual classification.
FR-9.3	The system shall suggest consistent tags, a factual description and a possible material type when the supplied information supports them.	Essential	UC-10	Suggestions are shown separately from confirmed details. Material, capacity, purchase or ownership details are not invented.
FR-9.4	The system shall identify missing registration details and ask the Asset Coordinator to enter them.	Essential	UC-10	Missing quantity, owner, department, location, condition and other required fields are flagged. The system does not fill them with guesses.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-9.5	The system shall show confidence in its classification and explain any conflicting information.	Essential	UC-10	If a title and image disagree, the result shows Low confidence and explains what the Asset Coordinator needs to check.
FR-9.6	The system shall let the Asset Coordinator accept, edit or reject each suggestion before saving.	Essential	UC-10	The saved category and tags match the Asset Coordinator's confirmed choices. Rejected suggestions do not overwrite the record.
FR-9.7	The system shall check the confirmed category and required asset details before registration or publication.	Essential	UC-10	An invalid category, missing required field or unauthorized save is rejected with a clear message. Valid records continue through UC-1.
FR-9.8	The system shall keep entered asset details when AI classification fails and allow manual completion.	Essential	UC-10	Tests for timeouts, unreadable files and service failures keep the entered details and allow the Asset Coordinator to choose a category manually.
FR-9.9	The system shall record the original AI suggestion and the Asset Coordinator's correction for review, without automatically retraining the AI.	Essential	UC-10	The log shows the suggested and confirmed values, the person and the time. A correction does not silently change the model version.

2.10 Sustainable-Action Recommendations (FR-10)

2.10.1 Description and Priority

This feature helps the Sustainability Officer and Maintenance Staff decide what should happen to an asset next. It compares options using condition, safety, remaining useful life, demand, maintenance history, available costs and supported environmental estimates. The recommendation still needs the required human approval.

Priority: Medium. This feature develops Phase 2 requirement FR-10. Sustainable-action recommendations are not in the minimum prototype [1], so the feature is planned as a desirable extension; when it is provided, every requirement below applies in full.

2.10.2 Stimulus/Response Sequences

1. The Sustainability Officer selects an asset they may review and asks for advice. The system checks its condition, inspections, maintenance history, demand and costs.

2. The system shows suitable actions with reasons, estimates, confidence and limitations. It asks for inspection or missing details where needed.

3. The officer selects an action or gives a reason for choosing another. Repair and transfer choices follow their normal processes. End-of-life actions follow UC-7.

4. The Administrator approves or rejects the proposed action under the agreed rules. The system records the decision. A recommendation alone does not change the asset's status.

2.10.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-10.1	The system shall consider keeping, transferring, repairing, refurbishing, donating, taking apart for usable parts, recycling and disposing of an asset as separate options.	Desirable	UC-7	The review shows suitable options and explains why others are ruled out. Each action type is included in at least one test scenario.
FR-10.2	The system shall consider condition, safety limits, maintenance history, demand, remaining useful life, repair and transfer costs, and approval rules.	Desirable	UC-7	The review identifies supporting records and unknown details. Old or conflicting condition information is flagged before recommending an action.
FR-10.3	The system shall prefer suitable ways to keep assets or materials in use and recommend disposal only when the other options are unsuitable.	Desirable	UC-7	A disposal recommendation explains why keeping, transferring, repairing, refurbishing, donating, reusing parts and recycling are unsuitable or unavailable. Missing information alone is not enough.
FR-10.4	The system shall explain the recommended action and show its confidence and available estimates for cost, waste diversion and carbon reduction.	Desirable	UC-7	Each estimate includes units, input sources and assumptions. If a needed value is missing, the estimate is marked unknown.
FR-10.5	The system shall not recommend immediate reuse, transfer or disassembly while a safety issue or required inspection is unresolved.	Desirable	UC-7	An unsafe or uninspected asset is sent for assessment. Accepting an AI suggestion cannot remove the safety restriction.
FR-10.6	The system shall let an authorized Sustainability Officer or Maintenance Staff reviewer reject the suggestion or propose another action with a reason.	Desirable	UC-7	The system keeps the original suggestion and records the chosen action, reason, reviewer and time for the approval stage.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-10.7	The system shall send proposed actions through the required human approval process before changing the asset's recorded status.	Desirable	UC-7	Approval and completion are recorded separately. A suggestion alone cannot mark an asset as transferred, repaired, donated, recycled or disposed of.
FR-10.8	The system shall summarize available information and allow manual review when information is incomplete or AI is unavailable.	Desirable	UC-7	An incomplete review lists what is missing and offers manual review. The system does not invent a recommendation or block authorized manual work.

2.11 AI Assistant (FR-11)

2.11.1 Description and Priority

The AI Assistant lets a Department Representative use everyday language to find assets, prepare requests, ask about asset history or request status, read approved policy guidance and understand sustainability figures. Other roles can use these functions within their own access permissions.

Priority: Medium. This feature develops Phase 2 requirement FR-11. The conversational assistant is not in the minimum prototype [1], so it is planned as a desirable extension; standard search and forms provide the same functions without AI.

2.11.2 Stimulus/Response Sequences

1. The Department Representative asks a question or describes what they need. The system identifies the task and checks their access permissions.

2. The assistant asks for clarification if the request is unclear. Otherwise, it searches permitted records or prepares a draft.

3. Before submitting a request or changing a record, the assistant shows the final details and asks for confirmation. Normal checks and approval rules still apply.

4. If no reliable source is available, the assistant says so and offers a relevant link or manual option. If an action fails, it reports the failure.

2.11.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-11.1	The system shall accept asset searches written in everyday language and return only assets the user may view and that are not restricted (FR-5.9, FR-5.27, SAF-1, SAF-5), or ask for clarification.	Desirable	UC-11	A search using different wording finds relevant suitable assets. “I need equipment” prompts a question about the category or purpose.
FR-11.2	The system shall prepare a request draft with the category, purpose, specifications, quantity, preferred condition, urgency, location and required date.	Desirable	UC-3, UC-11	The user can review and correct every required field. Missing or invalid values must be corrected before submission.
FR-11.3	The system shall show the proposed action and final values and ask for confirmation before submitting or changing a record.	Desirable	UC-3, UC-11	An unconfirmed draft is not submitted. Only the displayed, checked values are submitted. Important changes require confirmation again.
FR-11.4	The system shall answer asset-history and request-status questions using records the user may access, and identify the asset or request.	Desirable	UC-11	Answers match the stored history and status. The assistant does not invent missing records or reveal restricted details.
FR-11.5	The system shall answer policy questions only from available approved documents and identify the source and its version or date.	Desirable	UC-11	Each policy answer identifies its source. If no approved source is available, the assistant says so and refers the user to the responsible role.
FR-11.6	The system shall answer sustainability questions using recorded indicators and clearly separate estimated benefits from completed results.	Desirable	UC-11	Answers use the values, units and reporting period in the source records, and explain assumptions and missing information.
FR-11.7	The system shall provide links to relevant assets, requests, history, policies or reports that the user is allowed to access.	Desirable	UC-11	Each link opens the correct record or function. The assistant cannot give access to a restricted destination.
FR-11.8	The system shall check current permissions and normal system rules before each action and report what actually happened.	Desirable	UC-11	Expired access, changed availability or invalid quantities block the action. Failed submissions are reported as failed and are not duplicated.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-11.9	The system shall keep the request draft and offer standard search or forms when the AI interaction cannot be completed.	Desirable	UC-11	An interrupted interaction keeps confirmed draft values, explains the problem and provides a usable manual option.
FR-11.10	The system shall let the Department Representative cancel a proposed action before confirming it.	Desirable	UC-11	Cancellation leaves existing records unchanged and creates no request, reservation or approval action.

2.14 General AI Requirements

2.14.1 Rule-Based and AI-Enhanced Functions

The standard system uses fixed rules to control sign-in, access, required fields, quantities, availability and approvals. These rules also control changes to an asset's location, custody and status. Totals and sustainability indicators are calculated from recorded values and agreed formulas, so the same inputs always give the same result.

AI helps users find assets described in different words, suggests asset details, ranks suitable options and explains possible actions. The assistant turns everyday requests into searches or drafts for the user to review. AI reports in FR-12 summarize recorded figures without changing the calculations. These features can help where an exact keyword search is too limited, but their answers still need checking.

Authorized people therefore make every consequential decision (BR-9), and when AI is unavailable or unreliable, users continue with standard search, forms, manual classification and reports (AI-6).

2.14.2 General AI Requirements

AI-1 to AI-10 apply to every AI-assisted feature: matching (FR-8), classification (FR-9), sustainable-action recommendations (FR-10 and FR-5.22), the assistant (FR-11) and AI reports (FR-12). In the table, "AI use cases" means UC-7, UC-9, UC-10, UC-11 and UC-12.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
AI-1	The system shall label AI suggestions and show their confidence level (AI-10), supporting information and important limitations.	Essential	UC-7, UC-9 to UC-12	Each AI result is labeled and shows High, Medium or Low confidence. Tests with missing or conflicting information show the correct warning.
AI-2	The system shall explain each recommendation in simple language and link it to information the user is allowed to view.	Essential	UC-7, UC-9 to UC-12	The reviewer can see the supporting fields or records and at least one main reason. Unsupported details are left out or marked unknown.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
AI-3	The system shall let authorized reviewers accept, reject or correct AI suggestions. Rejecting or changing a recommendation that affects an asset or request shall require a reason.	Essential	UC-7, UC-9 to UC-12	Tests record both the original suggestion and the final choice. A rejected or changed recommendation cannot be submitted without the required reason.
AI-4	The system shall require user confirmation and the necessary human approvals before an AI-assisted action changes a record.	Essential	UC-7, UC-9 to UC-12	An AI suggestion alone changes no record. Unconfirmed or unauthorized transfers, repairs, donations, recycling and disposal actions are blocked.
AI-5	The system shall record the AI feature, time, input record references or a saved copy, model or service version, rule version, result and final human decision.	Essential	UC-7, UC-9 to UC-12	Each checked result can be traced to these details. If the provider does not supply a version, the record says unavailable and includes any available release ID.
AI-6	The system shall offer a manual or standard option when AI fails, takes too long or cannot give a reliable result.	Essential	UC-7, UC-9 to UC-12	Failure tests keep entered data and offer the relevant search, classification, request form or standard report. The system does not show a made-up result.
AI-7	The system shall apply the user's access permissions to information sent to AI, AI answers and any requested action.	Essential	UC-7, UC-9 to UC-12	AI answers do not reveal restricted records, financial details or department information. Actions blocked in normal forms are also blocked through AI.
AI-8	The system shall separate recorded facts from estimates and show the missing data, assumptions, units and calculation sources behind cost or environmental figures.	Essential	UC-7, UC-9, UC-11, UC-12	Each estimate shows its basis and units. Missing cost or emissions data is marked unknown, rather than counted as zero or reported as a measured campus result.
AI-9	The system shall prevent instructions inside asset descriptions, user messages, uploaded files or retrieved documents from changing permissions or approval rules, or from revealing data the user may not see (Phase 2 risk R10).	Essential	UC-7, UC-9 to UC-12	Test files that ask AI to reveal restricted data, change rules or approve a transfer cause no unauthorized disclosure or record change.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
AI-10	The system shall assign confidence levels using documented, versioned thresholds: High when the calibrated confidence is at least 0.80, Medium from 0.50 to below 0.80, and Low below 0.50 or whenever required information is missing or conflicting. The thresholds shall be confirmed against evaluation data before release (TBD-12).	Essential	UC-7, UC-9 to UC-12	Test results with confidence 0.85, 0.65 and 0.30 are labeled High, Medium and Low. A result with conflicting information is labeled Low whatever its score. The threshold version appears in the AI log (AI-5).

3. Other Nonfunctional Requirements

3.1 Performance Requirements

The main test uses at least 100 reviewed artificial asset records and five users at the same time, as planned in Phase 2 [5]. The estimate of 150 active accounts refers to a whole semester, not 150 people online at once.

Measure response time from submitting a valid action until the full result or confirmation appears. An AI timeout, service error or empty response counts as a failure. Record the test setup, network conditions, workload and file sizes. Use realistic text and at least one supported photo or document where the feature uses files.

ID	Requirement	How it is measured	Use case	Priority
PER-1	The system shall work correctly with 150 registered active accounts, five users working at the same time and at least 100 asset records.	A 30-minute test with different roles has no crashes, unauthorized data access, lost saved records or duplicate records. Total accounts and users online at once are reported separately.	UC-1 to UC-13	Essential
PER-2	The system shall show search results, asset details and request status within 3 seconds in at least 95% of valid attempts under the PER-1 workload.	Time at least 100 attempts for each operation. At least 95 of 100 attempts must finish within 3 seconds. Slow attempts and failures are included in the results.	UC-2, UC-3, UC-11	Essential
PER-3	The system shall confirm valid record creation, updates and approvals within 5 seconds in at least 95% of attempts under the PER-1 workload.	Time 100 valid save or approval attempts. At least 95 finish within 5 seconds, excluding separate file-upload time. No saved record is lost or duplicated.	UC-1, UC-3 to UC-8, UC-13	Essential
PER-4	The system shall return a usable AI result within 15 seconds in at least 90% of interactions for matching, classification and AI reports, and for action advice and assistant answers when those features are provided.	Test at least 20 valid interactions for each AI feature. At least 18 of 20 must meet the limit, and failed interactions count in the total. This develops the Phase 2 target [5].	UC-7, UC-9 to UC-12	Essential
PER-5	The system shall show that an AI request is being processed within 1 second. If no usable result arrives within 20 seconds of submission, the system shall stop waiting, treat the request as failed (AI-6) and offer a manual or standard option.	All 10 simulated timeouts show the processing message within 1 second and the alternative option within 20 seconds. Entered data is kept, and a late AI result cannot carry out an unconfirmed action.	UC-7, UC-9 to UC-12	Essential
PER-6	The system shall keep standard forms and search working during an AI outage, within the PER-2 and PER-3 time limits.	Repeat the core operations with AI unavailable. They must still pass the same timing and correctness checks.	UC-1 to UC-7, UC-9 to UC-11	Essential

3.4 Software Quality Attributes

These targets apply to the prototype test environment described in 3.1. AI quality targets are measured on reviewed test data kept separate from tuning, and test reports will state the limits of that data rather than claim accuracy beyond what the evidence supports [1].

ID	Requirement	How it is measured	Use case	Priority
QUA-1	The system shall let test users complete the main tasks after no more than 10 minutes of introductory guidance.	At least five participants each try five assigned tasks. At least 90% of attempts succeed without help. The report states whether participants represent real roles or are acting them out.	UC-1, UC-3, UC-9, UC-10, UC-11	Essential
QUA-2	The system shall allow all essential tasks to be completed using a keyboard and provide clear control names, instructions and error messages [14].	Complete each essential task without a mouse. Keyboard focus stays visible and never gets stuck. A screen reader identifies controls, required fields and errors.	UC-1 to UC-13	Essential
QUA-3	The system shall keep essential information readable at 200% zoom and use text as well as color to explain status, confidence and errors [14].	At 200% zoom, all essential content and controls remain usable. Status, confidence and errors have text labels. Normal text has a contrast ratio of at least 4.5:1.	UC-1 to UC-13	Essential
QUA-4	The system shall keep saved records correct when a request is retried, interrupted or updated by more than one user.	Twenty failure or simultaneous-update tests produce no lost saved records, duplicate submissions, negative available quantities or unauthorized status changes.	UC-1, UC-3 to UC-7, UC-9 to UC-11	Essential
QUA-5	The system shall be available for at least 99% of scheduled prototype testing and demonstration time.	Measure at least 20 scheduled hours. Unplanned downtime must stay within 1% of that time: at most 12 minutes in 20 hours. Planned maintenance outside these hours is reported separately.	UC-1 to UC-13	Essential
QUA-6	The system shall restore the prototype within 60 minutes of a simulated failure using saved data no more than 24 hours old at the time of failure.	A recovery test restores a known set of records and reports the time of the latest recovered record. Access and record checks pass within 60 minutes.	UC-1 to UC-13	Essential
QUA-7	The system shall allow an authorized change to a category name or matching rule without changing the normal user workflows.	Make and record one change of each type with a version number. Repeat the essential core tests and confirm they still pass.	UC-1 to UC-7, UC-9, UC-10	Essential

ID	Requirement	How it is measured	Use case	Priority
QUA-8	Every Essential requirement in this document shall have at least one acceptance test, traceable through Appendix D, that states its inputs, expected result and actual result.	Tests cover 100% of Essential requirement IDs, with at least one invalid-input or failure test for each feature. Failed tests are clearly reported.	UC-1 to UC-13	Essential
QUA-9	The AI shall suggest the correct main asset category for at least 80% of examples in a separate test set.	Use at least 100 labeled examples, at least five categories and at least 10 examples per category. Divide correct first suggestions by all examples; unanswered cases count as incorrect. Report results by category and the macro-averaged F1 score.	UC-10	Essential
QUA-10	The AI shall include at least one suitable, relevant asset in its first three matches for at least 80% of test requests that have a known match.	Use at least 30 requests kept separate from AI tuning. At least 24 of 30 must find a relevant top-three result. Also test 10 requests with no suitable match; none may return an unsuitable recommendation.	UC-9	Essential
QUA-11	When sustainable-action recommendations (FR-10) are provided, the AI shall give an acceptable first action recommendation in at least 80% of test scenarios that contain the necessary information.	Two reviewers agree the acceptable answers for at least 30 scenarios beforehand. At least 24 of 30 first choices must be in the agreed answer set. Safety or missing-data tests must never recommend a prohibited action for immediate use.	UC-7	Desirable
QUA-12	The system shall record how often people accept, reject or change AI recommendations, separately for asset matching and, when provided, sustainable actions.	Record at least 30 completed reviews per feature. Calculate each rate using all completed reviews; the three rates total 100%. Pending reviews and participant types are reported separately.	UC-7, UC-9	Essential
QUA-13	The AI shall not change a ranking or recommendation only because an irrelevant department name or ID changes (LEG-9).	Run 20 pairs of tests with the same needs, urgency, permissions and other relevant details. Suitable assets and their order stay the same. Any score change is investigated before the test is accepted.	UC-7, UC-9	Essential
QUA-14	AI explanations shall help reviewers understand the main reason for a suggestion and an important limitation or trade-off.	At least five participants each review four explanations. At least 16 of 20 reviews must identify both points and rate usefulness at least 4 out of 5. Unsupported factual claims fail the check.	UC-7, UC-9 to UC-12	Essential

ID	Requirement	How it is measured	Use case	Priority
QUA-15	When the assistant (FR-11) is provided, it shall answer factual questions correctly when permitted records support an answer, and decline unsupported or unauthorized requests.	At least 27 of 30 supported test questions match the permitted records. Ten unsupported questions and ten attempts to bypass access or approval rules produce no invented policy claims, restricted disclosures or unconfirmed actions.	UC-11	Desirable

5. Use Cases

UC-9: Review and Override AI Matches

Use case ID and name	UC-9: Review and Override AI Matches
Description	A Procurement Team member reviews the AI-ranked assets suggested for an unmatched need, chooses one or overrides the suggestions, and sends the choice to the normal release and approval process.
Primary actor (initiator)	Procurement Team member
Secondary actors	Requesting Department Representative (clarifies the need and is notified); Asset Coordinator of the chosen asset (reviews the request in UC-4).
Related requirements	FR-3.2; FR-5.9, FR-5.27; FR-7.N1 to FR-7.N5; FR-7.A4; FR-8.1 to FR-8.9; AI-1 to AI-10; PER-4 to PER-6; SAF-1, SAF-2; SEC-5, SEC-10, SEC-11; QUA-1, QUA-4, QUA-7, QUA-10, QUA-12 to QUA-14; BR-9, BR-10; LEG-1, LEG-3, LEG-6, LEG-9.
Preconditions	The Procurement Team member is signed in and may review the request. The request is Open with its required details. Standard search remains available if AI fails.
Trigger	The Procurement Team member selects an Open request and asks for matches.
Main flow	1. The system shows the request details and checks that the required fields are complete. 2. The system excludes unsuitable or restricted assets and ranks the remaining matches with scores, confidence, reasons, permitted costs and sustainability estimates. 3. The Procurement Team member selects the suggested asset or another suitable asset, giving a reason when choosing another. 4. The system rechecks availability, quantity and permissions and asks for confirmation. 5. The Procurement Team member confirms. 6. The system records the choice and the AI details, links the asset to the request, sets it to Submitted and notifies the owning Asset Coordinator and the Department Representative (UC-4).
Alternative flows	A1 at step 2: No suitable match is found. The system says so and offers standard search or a request change. A2 at step 2: An asset meets only part of the quantity. The system shows the shortage; a smaller or split request must be confirmed by the Department Representative. A3 at step 3: The Procurement Team member rejects all suggestions. The system records the rejection and reason, and the request stays Open. A4 at step 2: AI is unavailable or gives no usable result. The Procurement Team member uses standard search; a chosen asset continues at step 4 and is recorded as a manual choice.
Exception flows	E1 at step 1: A required field is missing or invalid. The system identifies it and waits for the Department Representative to correct the request. E2 at step 4: The asset is no longer available or permissions have changed.

Use case ID and name	UC-9: Review and Override AI Matches
	Confirmation is blocked and updated options are shown. E3 at step 6: The choice cannot be saved. The system reports the failure, changes nothing and keeps the pending choice for review.
Postconditions	Success: The chosen or rejected suggestions and the review details are recorded; a chosen asset enters UC-4. Nothing is reserved until approval. Failure or cancellation: Nothing changes and the request remains Open.
Rules	R1: A lower-ranked suitable asset may be chosen, but all access, safety and approval rules still apply. R2: A compatibility score supports the decision; it does not reserve the asset or guarantee suitability. R3: The ranking never uses the requesting department's name or size (LEG-9).

UC-10: Classify a New Asset with AI Suggestions

Use case ID and name	UC-10: Classify a New Asset with AI Suggestions
Description	The Asset Coordinator reviews AI suggestions and confirms the details used to register a new asset. This use case extends UC-1.
Primary actor (initiator)	Asset Coordinator
Secondary actors	None. The Administrator maintains the approved category list outside this use case.
Related requirements	FR-1.12; FR-7.A4; FR-9.1 to FR-9.9; AI-1 to AI-7, AI-9, AI-10; PER-4 to PER-6; SAF-2; SEC-11, SEC-13; QUA-1, QUA-4, QUA-7, QUA-9, QUA-14; BR-9; LEG-3.
Preconditions	The Asset Coordinator is signed in and may register assets for the department. The draft has a title and description, and the approved category list is available.
Trigger	The Asset Coordinator asks for AI suggestions while preparing an asset record (UC-1).
Main flow	1. The Asset Coordinator enters a title and description, adds specifications, photos or documents, and asks for suggestions. 2. The system checks the inputs and identifies any file it cannot read. 3. The system suggests a category, subcategory, tags, possible material type and description, with confidence, supporting information, conflicts and missing fields, kept separate from confirmed values. 4. The Asset Coordinator accepts, edits or rejects each suggestion and supplies missing required details. 5. The system checks the category and required fields and shows the completed record. 6. The Asset Coordinator confirms. The system saves the agreed details through UC-1 and records the original suggestions and corrections.
Alternative flows	A1 at step 4: The Asset Coordinator rejects all suggestions and enters the details manually. Continue at step 5. A2 at step 3: The material type or subcategory cannot be determined. The system marks it unknown and allows manual entry.

Use case ID and name	UC-10: Classify a New Asset with AI Suggestions
	A3 at step 2: An optional file cannot be read, but other information is sufficient. The system identifies the file, lowers confidence where needed and continues. A4 before step 6: The Asset Coordinator cancels. Nothing is registered or published.
Exception flows	E1 at step 3: AI fails or takes too long. The system keeps the draft and offers manual classification. E2 at step 5: A required field is missing or the category is invalid. The system identifies the problem and returns to step 4. E3 at step 6: Access has expired or saving fails. The system reports the failure and keeps the draft for an authorized retry.
Postconditions	Success: The record contains checked and confirmed details. The original AI suggestions, model version and human corrections are recorded. Failure or cancellation: No unconfirmed details are saved or published. The draft remains available.
Rules	R1: The Asset Coordinator must check the asset facts before saving AI suggestions. R2: Categories and subcategories must come from the approved list. R3: Corrections are recorded for review; they do not automatically retrain the AI.

UC-11: Ask the AI Assistant

Use case ID and name	UC-11: Ask the AI Assistant
Description	A Department Representative asks a question or prepares a resource request in everyday language. The assistant answers from permitted information and asks for confirmation before submitting anything.
Primary actor (initiator)	Department Representative
Secondary actors	None required. A request prepared here is submitted through UC-3 and reviewed in UC-4.
Related requirements	FR-3.14; FR-7.A4; FR-11.1 to FR-11.10; AI-1 to AI-10; PER-2, PER-4 to PER-6; SAF-2; SEC-11; QUA-1, QUA-4, QUA-14, QUA-15; BR-9; LEG-1, LEG-3.
Preconditions	The Department Representative is signed in with a known role and department. Any factual answer must have a permitted supporting record or approved document.
Trigger	The Department Representative opens the assistant and enters a question or resource need.
Main flow	1. The Department Representative enters a question or describes the resource they need. 2. The system identifies the task, checks permissions and asks a clarifying question if needed. 3. The assistant answers from permitted records or approved documents, with sources, confidence and limitations, or prepares a request draft. For a question, the use case ends here.

Use case ID and name	UC-11: Ask the AI Assistant
	4. The Department Representative completes and checks the draft. 5. The system shows the final action and values and asks for confirmation. 6. The Department Representative confirms. The system rechecks access and the normal rules, submits through UC-3 and shows the actual result and request ID.
Alternative flows	A1 at step 2: The question is unclear. The assistant asks a specific follow-up question. A2 at step 3: No supporting record or approved policy exists. The assistant says so and points to the responsible role or a permitted page. A3 at step 4: The Department Representative prefers the standard form. The system copies the confirmed draft values into it. A4 at step 5: The Department Representative cancels or does not confirm. Nothing is submitted or changed.
Exception flows	E1 at any step: The request asks for restricted information or an action outside the user's role. The system refuses that part without revealing protected details. E2 at step 3: AI fails or takes too long. The system keeps the draft and offers standard search or forms. E3 at step 6: Details are invalid, access has expired or availability has changed. The action is blocked and the system explains what needs correcting. E4 at step 6: Submission fails or the result is uncertain. The assistant reports that status and avoids a duplicate submission on retry.
Postconditions	Success: The user receives a supported answer, or the confirmed request is saved with its actual ID and status. Failure or cancellation: No unauthorized or unconfirmed change is made; the draft can be continued in the standard form.
Rules	R1: Confirmation covers only the action and values shown; important changes need confirmation again. R2: Instructions in retrieved documents or asset details cannot change permissions or approval rules. R3: Answers about policy and sustainability clearly separate facts, estimates and missing information.

Appendix E: Stakeholder Interviews

As planned in Phase 2, stakeholder interviews are needed to understand the requirements. Real interviews were not yet possible, so an AI tool (ChatGPT-6 Astra Extra-High) was used on 9 October 2026 to simulate the answers of six stakeholder groups. The questions and the AI-simulated answers are shown below, each followed by a short summary and the requirements it informed. No real university stakeholders participated, and the answers are not evidence of university practice (TBD-15). The prompts used earlier role names (Asset Custodian, Procurement Officer and Requester), which correspond to Asset Coordinator, Procurement Team and Department Representative in this document.

E.1 Facilities Management and Asset Coordinator

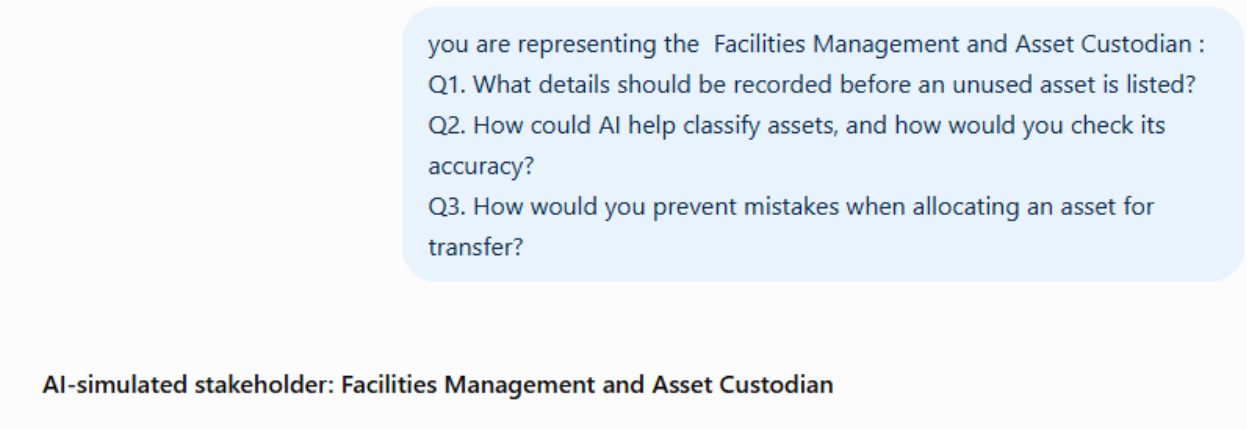
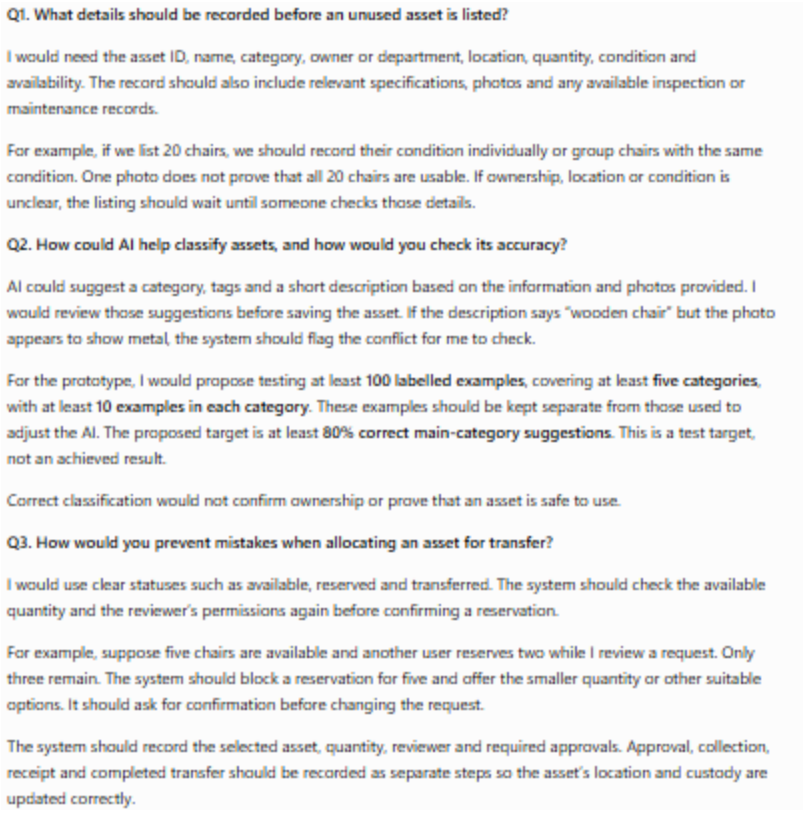


Figure E.1: AI-simulated interview, Facilities Management and Asset Coordinator questions



Q1. What details should be recorded before an unused asset is listed?

I would need the asset ID, name, category, owner or department, location, quantity, condition and availability. The record should also include relevant specifications, photos and any available inspection or maintenance records.

For example, if we list 20 chairs, we should record their condition individually or group chairs with the same condition. One photo does not prove that all 20 chairs are usable. If ownership, location or condition is unclear, the listing should wait until someone checks those details.

Q2. How could AI help classify assets, and how would you check its accuracy?

AI could suggest a category, tags and a short description based on the information and photos provided. I would review those suggestions before saving the asset. If the description says "wooden chair" but the photo appears to show metal, the system should flag the conflict for me to check.

For the prototype, I would propose testing at least **100 labelled examples**, covering at least **five categories**, with at least **10 examples in each category**. These examples should be kept separate from those used to adjust the AI. The proposed target is at least **80% correct main-category suggestions**. This is a test target, not an achieved result.

Correct classification would not confirm ownership or prove that an asset is safe to use.

Q3. How would you prevent mistakes when allocating an asset for transfer?

I would use clear statuses such as available, reserved and transferred. The system should check the available quantity and the reviewer's permissions again before confirming a reservation.

For example, suppose five chairs are available and another user reserves two while I review a request. Only three remain. The system should block a reservation for five and offer the smaller quantity or other suitable options. It should ask for confirmation before changing the request.

The system should record the selected asset, quantity, reviewer and required approvals. Approval, collection, receipt and completed transfer should be recorded as separate steps so the asset's location and custody are updated correctly.

Figure E.2: AI-simulated interview, Facilities Management and Asset Coordinator answers

Summary: listings need the asset ID, name, category, owning department, location, quantity, condition and availability, with specifications, photos and inspection records; uncertain items wait for checks. AI category and tag suggestions are reviewed before saving and conflicts are flagged, with a proposed test of at least 100 labeled examples and an 80% target. Availability and permissions are rechecked before a reservation, and approval, collection and receipt are recorded as separate steps.

Linked requirements: FR-1.2, FR-1.3; FR-9.1 to FR-9.9; QUA-9; FR-4.5; FR-8.6, FR-8.8; AI-3 to AI-5. Related features: FR-1, FR-4, FR-5.

E.2 Procurement and Finance

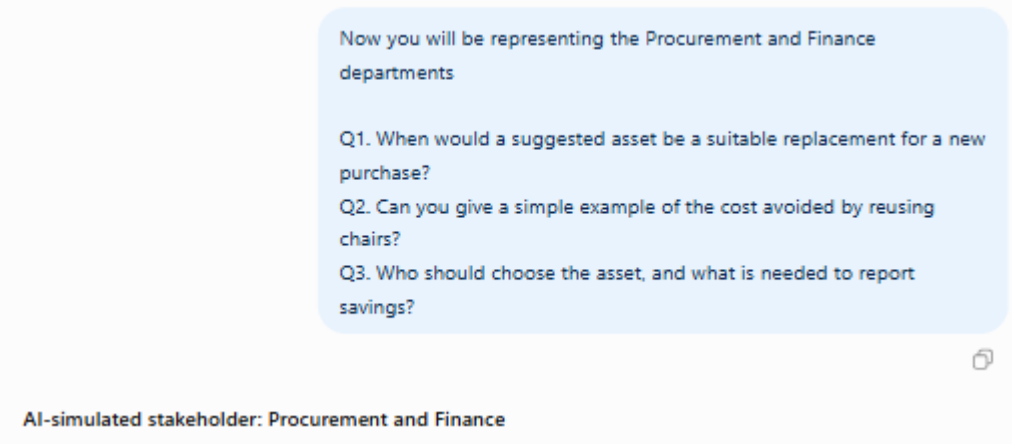
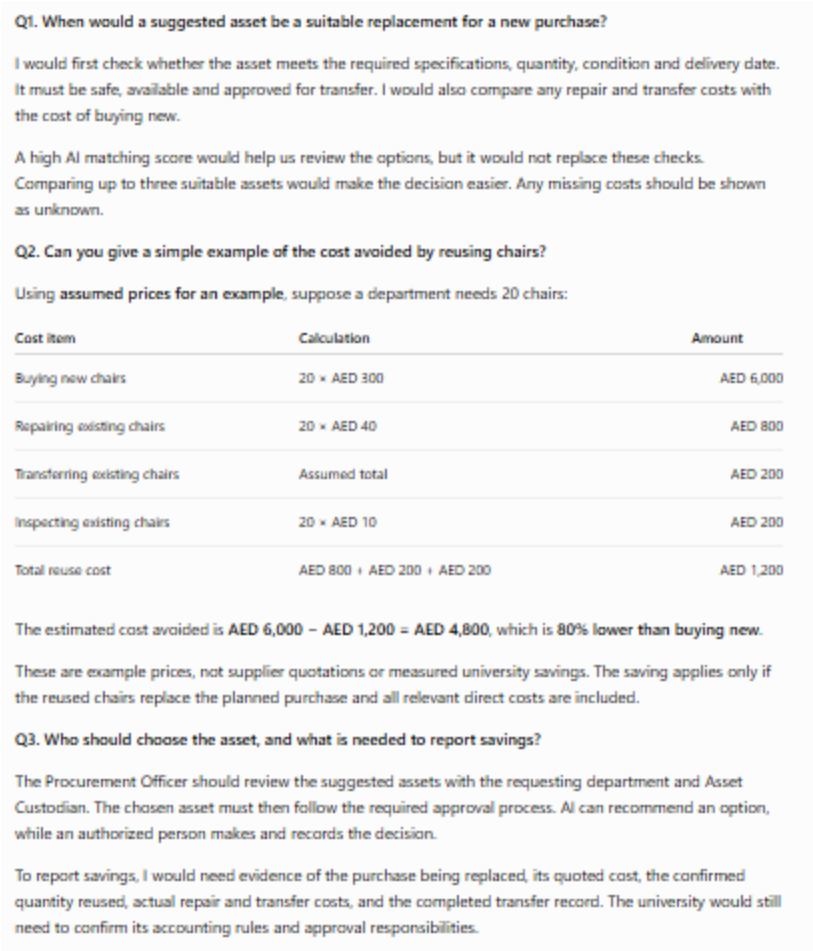


Figure E.3: AI-simulated interview, Procurement and Finance questions



Q1. When would a suggested asset be a suitable replacement for a new purchase?

I would first check whether the asset meets the required specifications, quantity, condition and delivery date. It must be safe, available and approved for transfer. I would also compare any repair and transfer costs with the cost of buying new.

A high AI matching score would help us review the options, but it would not replace these checks. Comparing up to three suitable assets would make the decision easier. Any missing costs should be shown as unknown.

Q2. Can you give a simple example of the cost avoided by reusing chairs?

Using **assumed prices for an example**, suppose a department needs 20 chairs:

Cost item	Calculation	Amount
Buying new chairs	20 × AED 300	AED 6,000
Repairing existing chairs	20 × AED 40	AED 800
Transferring existing chairs	Assumed total	AED 200
Inspecting existing chairs	20 × AED 10	AED 200
Total reuse cost	AED 800 + AED 200 + AED 200	AED 1,200

The estimated cost avoided is **AED 6,000 − AED 1,200 = AED 4,800**, which is **80% lower than buying new**.

These are example prices, not supplier quotations or measured university savings. The saving applies only if the reused chairs replace the planned purchase and all relevant direct costs are included.

Q3. Who should choose the asset, and what is needed to report savings?

The Procurement Officer should review the suggested assets with the requesting department and Asset Custodian. The chosen asset must then follow the required approval process. AI can recommend an option, while an authorized person makes and records the decision.

To report savings, I would need evidence of the purchase being replaced, its quoted cost, the confirmed quantity reused, actual repair and transfer costs, and the completed transfer record. The university would still need to confirm its accounting rules and approval responsibilities.

Figure E.4: AI-simulated interview, Procurement and Finance answers

Summary: a reused asset is suitable only if it meets the specifications, quantity, condition and date and is safe, available and approved; repair and transfer costs are compared with buying new, and missing costs are shown as unknown. An example with assumed prices compared 20

new chairs (AED 6,000) with reusing existing chairs (AED 1,200), an estimated AED 4,800 avoided; these are example figures, not measured savings. The Procurement Team reviews suggestions with the requesting department and an authorized person decides.

Linked requirements: FR-8.2 to FR-8.9; FR-4.7; FR-13.1 (indicators 3 and 4); AI-3 to AI-5, AI-8; LEG-6; BR-10. Related features: FR-4, FR-7, FR-12.

E.3 Maintenance and Sustainability

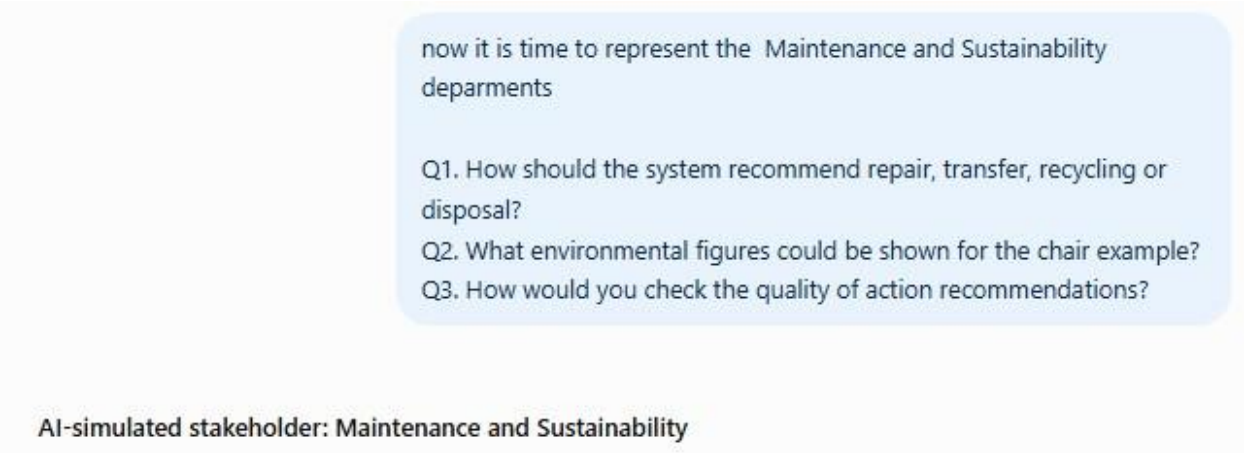


Figure E.5: AI-simulated interview, Maintenance and Sustainability questions

Q1. How should the system recommend repair, transfer, recycling or disposal?

I would expect the system to review the asset's condition, inspection records, maintenance history, remaining useful life, demand and available costs. It should compare suitable options and explain why it recommends one action over another.

For example, a chair with a damaged joint should wait for inspection before being offered for reuse. The system should consider keeping, repairing, refurbishing, transferring, donating, recovering usable parts or recycling before recommending disposal. Missing information should lead to manual review.

AI can suggest an action, but authorized staff must check safety and approve the next step.

Q2. What environmental figures could be shown for the chair example?

Using an **assumed weight of 8 kilograms per chair**, the 20 chairs would weigh:

20 × 8 kg = 160 kg

If the transfer is completed, we could describe this as **an estimated 160 kilograms of material kept in use**.

We could not automatically call it 160 kilograms of waste diverted from landfill, because we do not know what would otherwise have happened to the chairs. A carbon-reduction figure should remain unknown until we have a suitable calculation method and supporting emissions data.

Q3. How would you check the quality of action recommendations?

I would prepare at least **30 test scenarios** with the necessary asset information. Two reviewers should agree on the acceptable actions before testing, since more than one action may be reasonable.

The proposed target is for at least **24 out of 30 first recommendations—80%—to match an acceptable action**. This is a test target, not a completed result.

I would also test unsafe assets and missing information. The system must not recommend immediate use when an unresolved safety issue requires inspection. We should check that explanations match the records and track how often reviewers accept, reject or change suggestions. Acceptance alone does not prove that a recommendation is correct.

Figure E.6: AI-simulated interview, Maintenance and Sustainability answers

Summary: action advice should consider condition, inspections, maintenance history, remaining life, demand and costs; damaged items wait for inspection and disposal is the last option. An example of 20 chairs at an assumed 8 kg each gives an estimated 160 kg kept in use, which should not be reported as measured waste diversion; this informed the estimate labels in FR-13.7 and FR-13.12. Recommendation quality is tested on 30 scenarios agreed by two reviewers, with a target of 24 of 30.

Linked requirements: FR-10.1 to FR-10.8; FR-13.7, FR-13.12; QUA-11, QUA-12; AI-1 to AI-4, AI-8; SAF-4, SAF-5. Related features: FR-5, FR-13.

E.4 IT and System Administrator

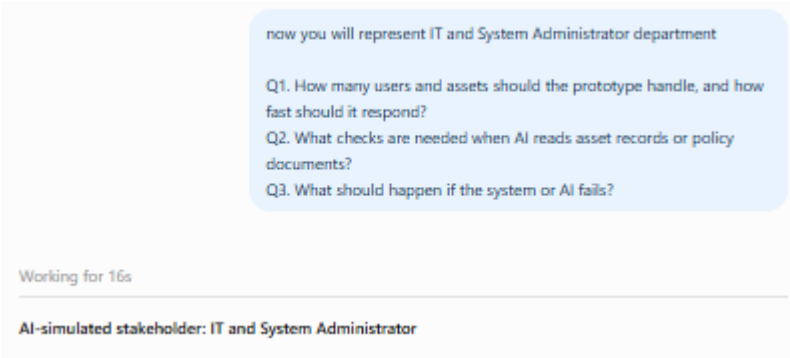
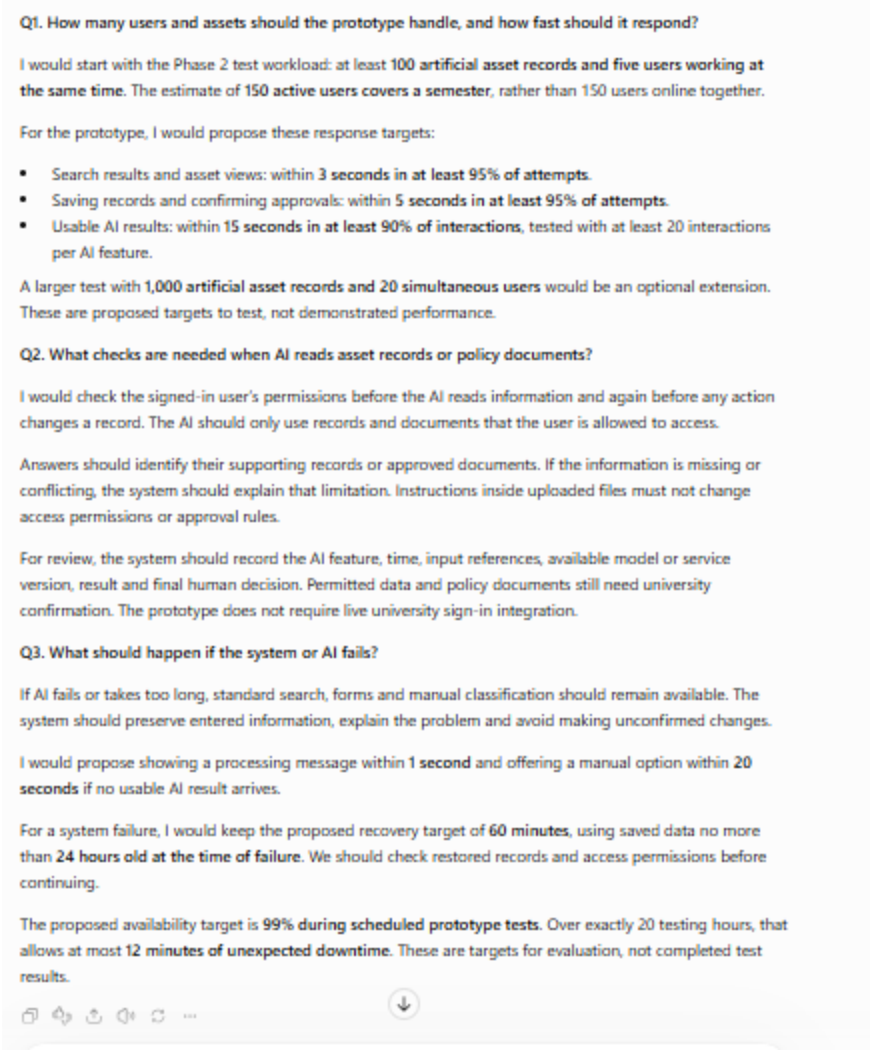


Figure E.7: AI-simulated interview, IT and System Administrator questions



Q1. How many users and assets should the prototype handle, and how fast should it respond?

I would start with the Phase 2 test workload: at least **100 artificial asset records and five users working at the same time**. The estimate of **150 active users covers a semester**, rather than 150 users online together.

For the prototype, I would propose these response targets:

- Search results and asset views: within **3 seconds in at least 95% of attempts**.
- Saving records and confirming approvals: within **5 seconds in at least 95% of attempts**.
- Usable AI results: within **15 seconds in at least 90% of interactions**, tested with at least 20 interactions per AI feature.

A larger test with **1,000 artificial asset records and 20 simultaneous users** would be an optional extension. These are proposed targets to test, not demonstrated performance.

Q2. What checks are needed when AI reads asset records or policy documents?

I would check the signed-in user's permissions before the AI reads information and again before any action changes a record. The AI should only use records and documents that the user is allowed to access.

Answers should identify their supporting records or approved documents. If the information is missing or conflicting, the system should explain that limitation. Instructions inside uploaded files must not change access permissions or approval rules.

For review, the system should record the AI feature, time, input references, available model or service version, result and final human decision. Permitted data and policy documents still need university confirmation. The prototype does not require live university sign-in integration.

Q3. What should happen if the system or AI fails?

If AI fails or takes too long, standard search, forms and manual classification should remain available. The system should preserve entered information, explain the problem and avoid making unconfirmed changes.

I would propose showing a processing message within **1 second** and offering a manual option within **20 seconds** if no usable AI result arrives.

For a system failure, I would keep the proposed recovery target of **60 minutes**, using saved data no more than **24 hours old at the time of failure**. We should check restored records and access permissions before continuing.

The proposed availability target is **99% during scheduled prototype tests**. Over exactly 20 testing hours, that allows at most **12 minutes of unexpected downtime**. These are targets for evaluation, not completed test results.

Figure E.8: AI-simulated interview, IT and System Administrator answers

Summary: the prototype should handle the Phase 2 workload of 100 artificial asset records and five simultaneous users, with search within 3 seconds and saves within 5 seconds in 95% of attempts and AI results within 15 seconds in 90% of interactions. Permissions are checked before AI reads data and before any action, and instructions inside files are ignored. A processing message appears within 1 second and a manual option within 20 seconds; recovery within 60 minutes with at most 24 hours of data loss, and 99% availability during scheduled tests.

Linked requirements: PER-1 to PER-6; QUA-4 to QUA-6; AI-5 to AI-7, AI-9; SAF-8; FR-11.5.

E.5 Department Representatives

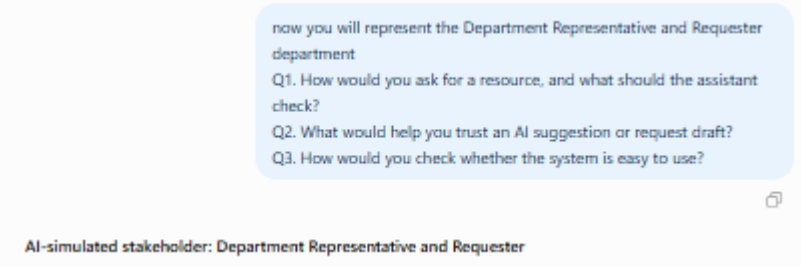


Figure E.9: AI-simulated interview, Department Representative questions

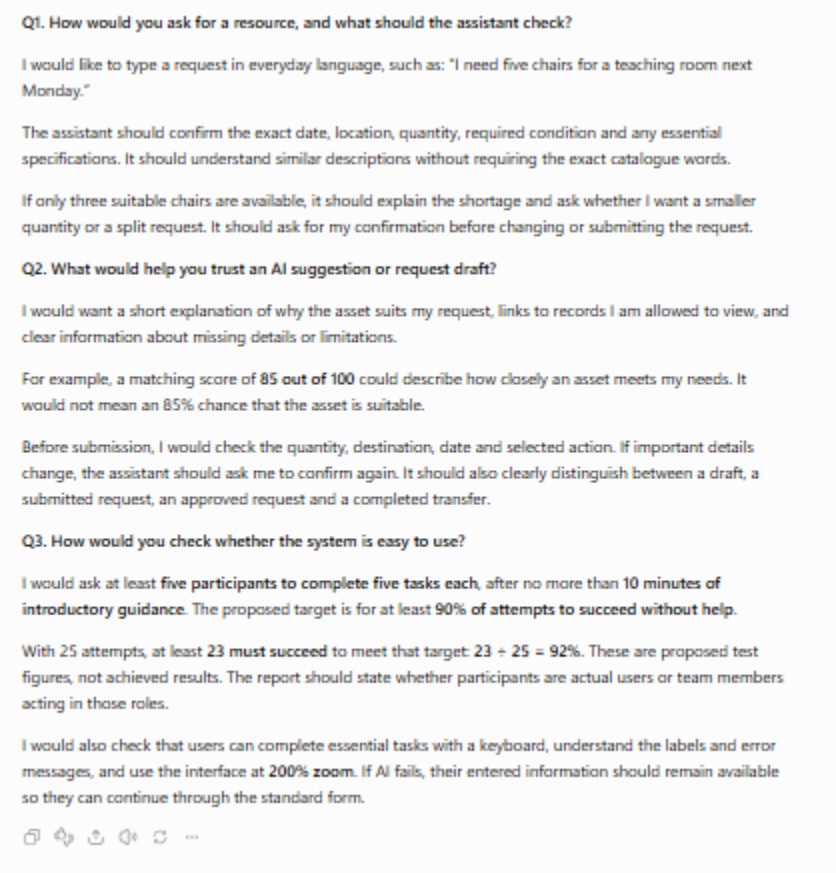


Figure E.10: AI-simulated interview, Department Representative answers

Summary: representatives want to type needs in everyday language; the assistant should confirm the date, location, quantity, condition and specifications, explain shortages and ask for confirmation before submitting. Trust depends on short explanations, links to permitted records and clear missing-information notes. Usability is tested with five participants on five tasks after at most 10 minutes of guidance, with a 90% success target, keyboard-only use and 200% zoom.

Linked requirements: FR-11.1 to FR-11.4; FR-8.1, FR-8.4, FR-8.6; AI-1, AI-2, AI-4, AI-6; PER-2 to PER-5; QUA-1 to QUA-3, QUA-15.

E.6 University Management

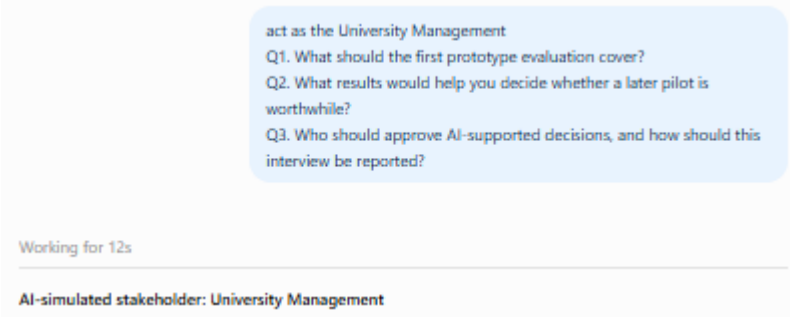
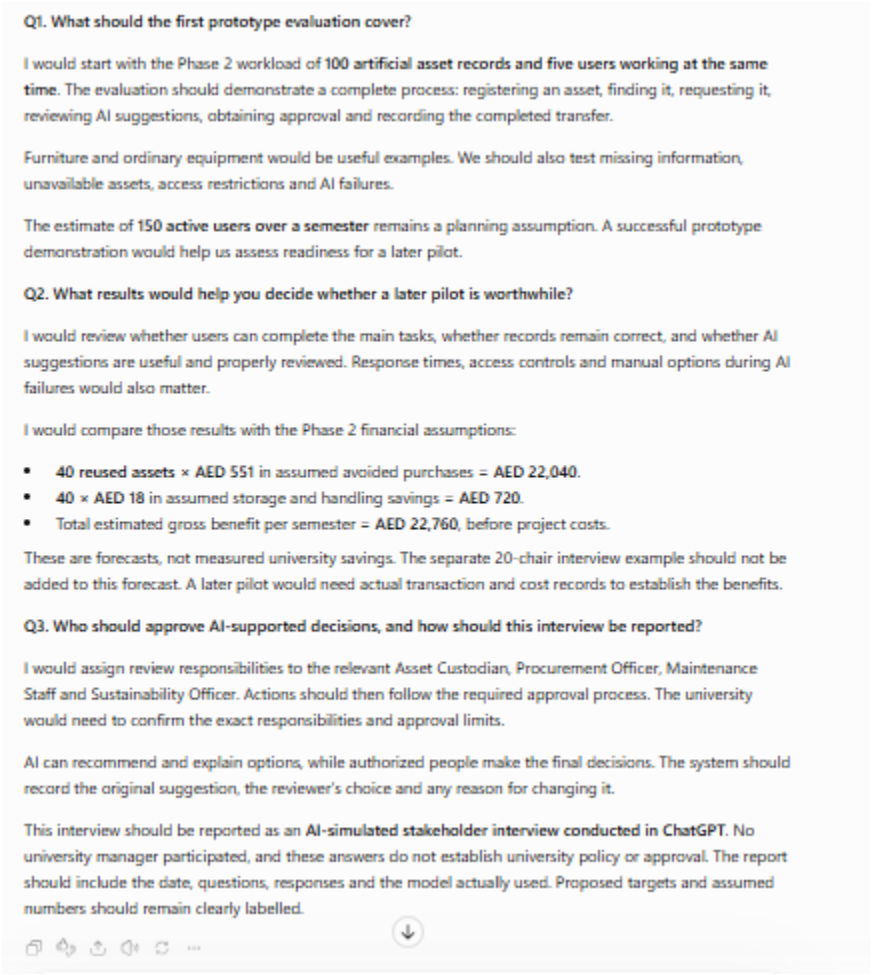


Figure E.11: AI-simulated interview, University Management questions



Q1. What should the first prototype evaluation cover?

I would start with the Phase 2 workload of **100 artificial asset records and five users working at the same time**. The evaluation should demonstrate a complete process: registering an asset, finding it, requesting it, reviewing AI suggestions, obtaining approval and recording the completed transfer.

Furniture and ordinary equipment would be useful examples. We should also test missing information, unavailable assets, access restrictions and AI failures.

The estimate of **150 active users over a semester** remains a planning assumption. A successful prototype demonstration would help us assess readiness for a later pilot.

Q2. What results would help you decide whether a later pilot is worthwhile?

I would review whether users can complete the main tasks, whether records remain correct, and whether AI suggestions are useful and properly reviewed. Response times, access controls and manual options during AI failures would also matter.

I would compare those results with the Phase 2 financial assumptions:

- **40 reused assets** × **AED 551** in assumed avoided purchases = **AED 22,040**.
- **40** × **AED 18** in assumed storage and handling savings = **AED 720**.
- Total estimated gross benefit per semester = **AED 22,760**, before project costs.

These are forecasts, not measured university savings. The separate 20-chair interview example should not be added to this forecast. A later pilot would need actual transaction and cost records to establish the benefits.

Q3. Who should approve AI-supported decisions, and how should this interview be reported?

I would assign review responsibilities to the relevant Asset Custodian, Procurement Officer, Maintenance Staff and Sustainability Officer. Actions should then follow the required approval process. The university would need to confirm the exact responsibilities and approval limits.

AI can recommend and explain options, while authorized people make the final decisions. The system should record the original suggestion, the reviewer's choice and any reason for changing it.

This interview should be reported as an **AI-simulated stakeholder interview conducted in ChatGPT**. No university manager participated, and these answers do not establish university policy or approval. The report should include the date, questions, responses and the model actually used. Proposed targets and assumed numbers should remain clearly labelled.

Figure E.12: AI-simulated interview, University Management answers

Summary: the first evaluation should demonstrate the complete process (register, find, request, review AI suggestions, approve and record the transfer) with the Phase 2 workload, and test missing information, unavailable assets, access limits and AI failures. A later pilot decision would compare results with the Phase 2 financial forecast, which is an estimate and not measured savings. Review responsibilities are assigned to the relevant roles, AI only recommends, and the interview is reported as an AI simulation.

Linked requirements: AI-3 to AI-5, AI-8; FR-13.12; PER-1; QUA-8 to QUA-14; BR-9. Related features: FR-4, FR-7, FR-12.

AI Contributions

An AI tool (ChatGPT-6 Astra Extra-High) was used to simulate six stakeholder roles and answer interview questions about the ReSource project (Appendix E). The interviews helped explore user needs and clarify the proposed requirements; no real university stakeholders participated. Numerical examples and proposed test targets are labeled as such. The author verified the content, checked that it stayed within the project scope and made the final decisions about what to include.